



Exam :

End Term Quiz

Subject :

Maths2

Total Marks :

50.00

QP :

2024 Apr28: IIT M FN EXAM QDF2

Exam Mode

Learning Mode

QUESTION MENU

1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18
19	20	21	22	23	24	25	26	27
28	29	30	31	32	33	34	35	

TIMER

00:25



CONTROLS

✓ SUBMIT EXAM

Your Score

0.00 / 50.00
(0%)

Question 1 : 640653815389

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "FOUNDATION LEVEL: SEMESTER II: MATHEMATICS FOR DATA SCIENCE II (COMPUTER BASED EXAM)" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

OPTIONS :

☐ YES☐ NO

Your score : 0

Discussions (0)



Question 2 : 640653815394

View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Let $S = \{v_1, v_2, v_3, v_4\}$ be a subset of $\mathbb{R}^3$. Select all true statements.



OPTIONS :

☐ S is linearly dependent.☐ S is linearly independent.☐ Deleting any one element from S turns it into a basis.☐ $\text{span}(S) = \mathbb{R}^3$ 

Your score : 0

Discussions (0)



Question 3 : 640653815415

View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Let $f(x, y) = 2x^4 - 8\sqrt{y} - 7$. Choose the correct options from the following:



OPTIONS :

☐ f is increasing at the point $(1, 4)$, if y is fixed and x is varied.☐ f is decreasing at the point $(1, 4)$, if y is fixed and x is varied.☐ f is decreasing at the point $(1, 4)$, if x is fixed and y is varied.

☐ None of these

Your score : 0

 Discussions (0)



Question 4 : 640653815390

 View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Let A and B be two square matrices of the same order. Select all true statements.

OPTIONS :

- ☐ The i th column of AB is a linear combination of the columns of A.
- ☐ The i th row of AB is a linear combination of the rows of B.
- ☐ The i th column of AB is a linear combination of the columns of B.
- ☐ The i th row of AB is a linear combination of the rows of A.

Your score : 0

 Discussions (0)



Question 5 : 640653815395

 View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Consider the matrix $A = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix}$. If A is invertible, which of the following conditions must be satisfied? Select all conditions that apply.

OPTIONS :

☐ $a \neq b$ 

☐ $a \neq 0$ 

☐ $a = b$ 

☐ $a = b = 0$ 

☐ $a = 0$

Your score : 0

Discussions (0)



Question 6 : 640653815410

View Solutions (0)

Total Mark : 4.00 | Type : MSQ

Choose the correct option(s) from the following:

OPTIONS :

☐ The orthonormal set obtained from $\{(0, 1), (-1, 3)\}$ using Gram-Schmidt process is $\{(0, 1), (-1, 0)\}$.

☐ $\begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$ is an orthogonal matrix.

☐ Let $W = \{(x, y, z) \in \mathbb{R}^3 : x + 2y + z = 0\}$. If $u = (1, -1, 1)$ and if P_W is the projection from $\mathbb{R}^3$ to W , then $P_W(u) = u$.

☐ The determinant of a matrix formed by 3 orthonormal vectors in $\mathbb{R}^3$ is always 1.

Your score : 0

Discussions (0)



Question 7 : 640653815414

View Solutions (0)

Total Mark : 4.00 | Type : MSQ

Choose the option(s) for which the limit exists.

OPTIONS :

☐ $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^4 + y^2}$

☐ $\lim_{(x,y) \rightarrow (0,0)} \frac{|xy|}{x^2 + y^2}$

☐ $\lim_{(x,y) \rightarrow (\pi,1)} \frac{x}{y} - \sin xy$

☐ $\lim_{(x,y) \rightarrow (1,1)} \frac{5x^2 - 6xy + y^2}{x^2 - y^2}$

Your score : 0

Discussions (0)



Question 8 : 640653815391

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ and $y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ be two non-zero column vectors in $\mathbb{R}^n$.

Your score : 0



Question 9 : 640653815392

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : MCQ

What is the dimension of xy^T ?

OPTIONS :

☐ $n \times n$

☐ $1 \times n$

☐ $n \times 1$ 

☐ 1×1 

Your score : 0

 Discussions (0)



Question 10 :
640653815393

 View Parent QN

 View Solutions (0)

Total Mark : 1.00 | Type : SA

What is the rank of xy^T ? 

Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

 Discussions (0)



Question 11 : 640653815396

Total Mark : 0.00 | Type : COMPREHENSION

Let A be a 3×5 matrix. Based on the above data, answer the given subquestions.

Your score : 0



Question 12 :
640653815397

 View Parent QN

 View Solutions (0)

Total Mark : 1.00 | Type : MCQ

How many solutions does $Ax = 0$ have? 

OPTIONS :

- ☐ Exactly one solution
- ☐ Infinitely many solutions
- ☐ No solution
- ☐ Insufficient data. This depends on the entries of A.

Your score : 0

[Discussions \(0\)](#)

Question 13 :
640653815398

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : MCQ

How many solutions does $A^T x = 0$ have? 

OPTIONS :

- ☐ Exactly one solution
- ☐ Infinitely many solutions
- ☐ No solution
- ☐ Insufficient data. This depends on the entries of A.

Your score : 0

[Discussions \(0\)](#)

Question 14 :
640653815399

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : MCQ

Let $T : \mathbb{R}^5 \rightarrow \mathbb{R}^3$ be the linear transformation whose matrix representation is A with respect to the standard basis for both domain and co-domain.

Which of the following is true?

OPTIONS :

- ☐ T could be onto, but it can never be one-one
- ☐ T could be one-one, but it can never be onto
- ☐ T could be both one-one and onto

Your score : 0

 Discussions (0)



Question 15 : 640653815400

Total Mark : 0.00 | Type : COMPREHENSION

Find the dimension of the given vector spaces in the subquestions.

Your score : 0



Question 16 : 640653815401

 View Parent QN

 View Solutions (0)

Total Mark : 1.00 | Type : SA

$U = \{x \in \mathbb{R}^4 \mid \langle x, u \rangle = 0 \text{ where } u = (1, -1, 2, -1)\}$ is a subspace of $\mathbb{R}^4$ and the dot product is used as the inner product.

Answer (Numeric):

Answer

Accepted Answer : 3

Your score : 0

Discussions (0)



Question 17 :
640653815402

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : SA

$$V = \left\{ \begin{bmatrix} x & 0 & y \\ 0 & 0 & 0 \\ 0 & x+y & 0 \end{bmatrix} : x, y \in \mathbb{R} \right\},$$



where V is a subspace of
 $M_{3 \times 3}(\mathbb{R})$, the vector
space of all 3×3 real matrices.

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

Discussions (0)



Question 18 :
640653815403

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : SA

$$W = \{(x, y, z) : x + y - 2z = 0$$

and $2x - y + z = 0$ and $x, y, z \in \mathbb{R}\},$
where W is a subspace of $\mathbb{R}^3$.



Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

Discussions (0)



Question 19 : 640653815404

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation given by

$$T(x) = x \text{ for all } x \in \mathbb{R}^2.$$



Your score : 0

Question 20 :
640653815405

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : MCQ

Which of the following is the matrix representation of T with respect to the basis $\{(1, 1), (-1, 1)\}$ for the domain and the standard ordered basis for the co-domain?

OPTIONS :

☐ $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

Your score : 0

Discussions (0)

**Question 21 :**
640653815406

View Parent QN



View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Which of the following are valid matrix representations of T with suitable bases for the domain and co-domain? Note that the bases could be different for the domain and co-domain.

OPTIONS :

☐ $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ 

☐ $\begin{bmatrix} 6 & 3 \\ 4 & 2 \end{bmatrix}$ 

☐ $\begin{bmatrix} -1 & 4 \\ 3 & 0 \end{bmatrix}$ 

☐ $\begin{bmatrix} -2 & 1 \\ 2 & -1 \end{bmatrix}$ 

Your score : 0

Discussions (0)

**Question 22 : 640653815407**

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

If (a, b) is the point on the line $y = -x$ nearest to the point $(2, -4)$,



Your score : 0



Question 23 :
640653815408

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

find a.

Answer (Numeric):

Answer

Accepted Answer : 3

Your score : 0

Discussions (0)



Question 24 :
640653815409

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

find b.

Answer (Numeric):

Answer

Accepted Answer : -3

Your score : 0

Discussions (0)



Question 25 : 640653815411

Total Mark : 0.00 | Type : COMPREHENSION

Answer the given subquestions:

Your score : 0



Question 26 :
640653815412

View Parent QN

View Solutions (0)

Total Mark : 2.00 | Type : SA

Suppose the function $T(x, y, z) = \frac{xz}{x^2 + y^2}$

represents temperature at the point (x, y, z) in a room and (u_1, u_2, u_3) is the unit vector in the direction in which the temperature increases most rapidly at the point $(1, 0, 1)$.

Find $u_1 + u_2 + u_3$.

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

Discussions (0)



Question 27 :
640653815413

View Parent QN

View Solutions (0)

Total Mark : 2.00 | Type : SA

Suppose a is the rate of change
of the function $f(x, y, z) = x^3yz^2$
at the point $(-1, 2, 1)$ in
the direction where f decreases
most rapidly. Find a^2 .

Answer (Numeric):

Accepted Answer : 53

Your score : 0

 Discussions (0)

Question 28 : 640653815416

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let $f(x, y) = x^3 + y^3 - 3(x + y^2)$. 

Your score : 0

Question 29 :
640653815417 View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

How many critical points are there for f?

Answer (Numeric):

Accepted Answer : 4

Your score : 0

 Discussions (0)Question 30 :
640653815418 View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

How many saddle points are there for f ?

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

 Discussions (0)



Question 31 :

640653815419



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Choose the correct option(s) from the following:

OPTIONS :

- ☐ f has a local minimum at the point (1,2).
- ☐ f has a local maximum at the point (1,2).
- ☐ f has a local maximum at the point (1, 0).
- ☐ f has a local maximum at the point (-1, 0).

Your score : 0

 Discussions (0)



Question 32 : 640653815420

Total Mark : 0.00 | Type : COMPREHENSION

Answer the given subquestions about the functions

$$u(x, y) = xy + 2e^x \text{ and } v(x, y) = y^2 + ye^x$$



Your score : 0



Question 33 :
640653815421

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

If $L_u(x, y)$ is the linear approximation of the function $u(x, y)$ at point $(1, 2)$ and $L_u(2, 3) = a + be$, and if $L_v(x, y)$ is the linear approximation of the function $v(x, y)$ at point $(2, 1)$ and $L_v(2, 3) = c + de^2$, where a, b, c, d are integers, find $a + b$.

Answer (Numeric):

Accepted Answer : 9

Your score : 0

[Discussions \(0\)](#)

Question 34 :
640653815422

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

If $L_u(x, y)$ is the linear approximation of the function $u(x, y)$ at point $(1, 2)$ and $L_u(2, 3) = a + be$, and if $L_v(x, y)$ is the linear approximation of the function $v(x, y)$ at point $(2, 1)$ and $L_v(2, 3) = c + de^2$, where a, b, c, d are integers,

find $c + d$. 

Answer (Numeric):

Answer

Accepted Answer : 8

Your score : 0

 Discussions (0)



Question 35 :
640653815423

 View Parent QN

 View Solutions (0)

Total Mark : 4.00 | Type : MSQ

Choose the correct option(s) from the following:

OPTIONS :

☐ $z = 2y + (3 + 2e^2)x - 2e^2 - 6$ is the tangent plane to the function $u(x, y)$ at the point $(2, 3)$. 

☐ $x(t) = 2 + \frac{t}{\sqrt{2}}, y(t) = 3 + \frac{t}{\sqrt{2}}, z(t) = (6 + 2e^2) + \frac{t}{\sqrt{2}}(5 + 2e^2)$ is the tangent line of the function $u(x, y)$ at the point $(2, 3)$ in the direction of $(1, 1)$. 

☐ $z = 3e^2x + (6 + e^2)y - 6e^2 - 9$ is the tangent plane to the function $v(x, y)$ at the point $(2, 3)$. 

☐ $x(t) = 2 + \frac{t}{\sqrt{2}}, y(t) = 3 + \frac{t}{\sqrt{2}}, z(t) = (9 + 3e^2) + \frac{t}{\sqrt{2}}(3 + 2e^2)$ is the tangent line of the function $v(x, y)$ at the point $(2, 3)$ in the direction of $(1, 1)$. 

Your score : 0

 Discussions (0)



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